INC-2621: kafka degraded

Date: 2026-07-25

Severity: P3

Duration: 16 minutes

Services affected: kafka, and accounting, fraud-detection downstream

Responders: data (PD-DATA)

Summary

The `kafkaQueueProblems` fault was active for 16 minutes. Broker is overloaded and consumers are given an added delay, so lag spikes. `kafka` alerted directly.

Customer impact

No customer-facing impact. `accounting` and `fraud-detection` fell behind on the orders topic; ledger entries and fraud scores were delayed, not lost, and caught up after recovery.

Timeline

Time	Event
07:41	Fault injected: flag `kafkaQueueProblems` set to its fault variant. No alert yet.
07:43	First alert: `kafka` duration p95 crosses its page threshold. data and the data team are paged.
07:45	Responder confirms the alert on `kafka` and begins checking its own dependencies rather than `kafka` itself, since `kafka` code has not changed.
07:48	Trace inspection: a `kafka` request trace shows the time originating in the `kafka` span, not in `kafka`.
07:50	`kafka` RED panels match the known signature — kafka_consumer_group_lag_ratio climbing for the accounting and fraud-detection groups; checkout produce latency up; storefront unaffected.
07:52	flagd checked: `kafkaQueueProblems` is active, set at 07:41. Cause confirmed.
07:53	Flag reset to resting variant. `kafka` recovers on the next flag evaluation.
07:57	`kafka` metrics back at baseline for 3 minutes. Incident closed.

Root cause

The `kafkaQueueProblems` feature flag was set to its fault variant. Broker is overloaded and consumers are given an added delay, so lag spikes. `kafka` itself was healthy — correct version, no resource pressure, dependencies up. The failure was entirely injected at the flag layer.

Contributing factors

• No alert correlates flag changes with incident start. flagd emits `flagd_flag_evaluations_total` but nothing pages on a flip, so the 07:52 flagd check was manual and late.

What went well

• Once a trace was pulled (07:48), the cause was clear within two minutes.

• The `kafka` runbook's documented signature matched exactly, so no time was lost confirming.

Action items

1. Add a Grafana annotation and a low-urgency alert on `flagd_flag_evaluations_total` deltas, so a flag flip is visible on every dashboard. *(owner: platform-eng)*

2. Confirm the `kafka` runbook's known-failure-modes section lists `kafkaQueueProblems` with this signature. *(owner: data)*

Detection signature

```
kafka_consumer_group_lag_ratio climbing for the accounting and fraud-detection groups; checkout produce latency up; storefront unaffected.
```